

1. 證明數歸法

Let $S(n)$ denote an open statement

If (1) $S(1)$ is true

(2) if $S(k)$ is true then

$S(k+1)$ is also true

then $S(n)$ is true $n \in \mathbb{N}$

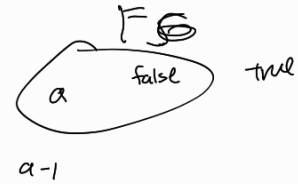
<pf>

Let $F = \{ t \in \mathbb{Z}^+ \mid S(t) \text{ is false} \}$

假設 $F \neq \emptyset$ ($F \subseteq \mathbb{Z}^+$, $F \neq \emptyset$)

By the well-ordering property

F has a least element a



$S(a)$ false

$S(1)$ true $a \neq 1$ $a > 1$

$\because a$ is "least" $\therefore (a-1) \notin F$ $S(a-1)$ true

$S((a-1)+1) = S(a)$

$\therefore (2) \therefore S(a)$ true $\Rightarrow *$

2.

證明: 任意大於 1 的合數 n 必存在一個質數 p , 使 p 整除 n

pf) 設 S 為所有沒有質因數合數的集合

$$\Rightarrow S = \begin{cases} S \subseteq \mathbb{Z}^+ \\ S \neq \emptyset \end{cases}$$

By the well-ordering property

設 S 中有一最小數 m

$$\because m = m_1 \cdot m_2 \quad (m_1, m_2 \text{ 有質因數的合數})$$

$$\text{且 } 1 < m_1, m_2 < m$$

$$\text{設 } p \text{ 為質數, 則 } p \mid m_1, m_1 \mid m \Rightarrow p \mid m \quad \times$$

3. 證明有無限多個質數

$$\text{Let } B = p_1 \times p_2 \times p_3 \times \dots \times p_k + 1$$

$$\Rightarrow B > p_i \quad (i = 1 \sim k)$$

$\therefore B$ is composite

設 p_j 為特定某數可整除 B

$$\Rightarrow p_j \mid B \quad - \textcircled{1}$$

$$p_j \mid p_1 \times p_2 \times p_3 \times \dots \times p_k + 1 \quad - \textcircled{2} \quad (\because \textcircled{1})$$

$$p_j \mid p_1 \times p_2 \times p_3 \times \dots \times p_k \quad - \textcircled{3} \quad (\because p_j \mid p_j)$$

$$p_j \mid 1$$

$$(\because \textcircled{2} \times 1 + \textcircled{3} \times (-1))$$

$$\Rightarrow p_j = \pm 1 \quad \times$$

4. 證明 GCD 最大公因數

$$\text{GCD}(12, 18) = 6$$

pf) Assume $a, b \in \mathbb{Z}$

$\exists c \in \mathbb{Z}^+$ is GCD

if (1) $c \mid a, c \mid b$

(2) for any common divisor d of a and b , we have $d \mid c$

5. Prove $a, b \in \mathbb{Z}^+$

there exist a unique $c \in \mathbb{Z}^+$ the gcd

<pf>

(I) Given $a, b \in \mathbb{Z}^+$

$$S = \{as + bt \mid s, t \in \mathbb{Z}\}$$

$$\Rightarrow as + bt > 0$$

$$S \begin{cases} S \neq \emptyset \\ S \subseteq \mathbb{Z}^+ \end{cases}$$

By the well-ordering property

S has the least element c

We claim c is the gcd

$$\Rightarrow c = ax + by$$

if $d|a, d|b$, then $d|ax+by \Rightarrow d|c$

(II) If $c \nmid a$ $a = qc + r$ $q, r \in \mathbb{Z}^+$

$$r = a - qc$$

$$0 < r < c$$

$$r = a - qc = a - q(ax + by) = (1 - qx)a + (-qy)b$$

$$\Rightarrow r \in S \quad \times \quad (\because r < c)$$

If $c \nmid b$ $b = pc + r'$ $p, r' \in \mathbb{Z}^+$

$$r' = b - pc$$

$$0 < r' < c$$

$$r' = b - pc = b - p(ax + by) = (-px)a + (1 - py)b$$

$$\Rightarrow r' \in S \quad \times \quad (\because r' < c)$$

(III) 唯一

c_1, c_2 satisfy (I) (II)

$$c_1 | c_2, c_2 | c_1 \Rightarrow c_1 = c_2$$

$$\left(\begin{array}{l} \because \textcircled{1} [a|b] \wedge [b|a] \Rightarrow a = \pm b \\ \textcircled{2} c_1, c_2 \in \mathbb{Z}^+ \end{array} \right)$$